

MULTIMODAL CoBIT: NATIVE MULTIMODAL GENERATION WITH BITSTREAM DIFFUSION

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ABSTRACT

[Luca will lead the pitch and framing of the paper; to be written last, once results are final.]

1 INTRODUCTION

[Luca will start working on the pitch of the paper here: motivation, the case for native multimodal generation with CoBit, and contributions.]

2 RELATED WORK

[To be written.]

3 METHOD

3.1 CONTINUOUS BITSTREAM DIFFUSION

[Recap of CoBit.]

3.2 MULTIMODAL BITSTREAMS

[How arbitrary modalities are encoded into a joint bitstream.]

3.3 GENERATIVE DIRECTIONS

[Unconditional, conditional and any-to-any generation via partial denoising.]

4 EXPERIMENTS

4.1 EXPERIMENTAL SETUP

[Shared setup common to all modalities.]

4.2 IMAGE GENERATION (CLASS-CONDITIONAL IMAGENET)

[Georgios to write: image-only results and comparisons.]

4.3 IMAGE AND TEXT (CC3M, CC12M)

[Georgios to write: image+text results and comparisons.]

4.4 AUDIO AND TEXT

[George to write: audio+text results and comparisons.]

054 4.5 PROTEIN SEQUENCE AND STRUCTURE

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056 *[Bruno to write: protein sequence and structure results and comparisons.]*
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058 5 DISCUSSION

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060 *[To be written.]*
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062 6 CONCLUSION

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064 *[To be written.]*
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067 REFERENCES

068 A IMAGE EXPERIMENTS

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070 *[Georgios: additional results and experimental details.]*
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073 B AUDIO AND TEXT EXPERIMENTS

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075 *[George: additional results and experimental details.]*
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078 C PROTEIN SEQUENCE AND STRUCTURE EXPERIMENTS

079 *[Bruno: additional results and experimental details for reproducibility.]*
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